TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	91.1 ± 1.04
Correlation Stability	94.0 ± 0.00
Distribution Consistency	93.4 ± 1.56
Entropy Preservation	85.2 ± 1.83
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	16.9 ± 2.66
Data Density	100.0 ± 0.00
Coverage Score	69.6 ± 4.22
Feature Accuracy	83.1 ± 2.55
Novelty Score	0.7 ± 0.02

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Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	3.064	-0.032	1.0%	OK
PRE4	3.282	3.316	+0.034	1.1%	OK
PRE5	4.569	5.825	+1.256	27.5%	Drift
PRE6	0.781	0.801	+0.020	2.5%	OK
PRE7	0.066	0.056	-0.010	15.3%	Warn
PRE8	0.145	0.114	-0.030	21.0%	Warn
PRE9	0.066	0.064	-0.002	3.2%	OK
PRE10	0.687	0.697	+0.010	1.4%	OK
PRE11	0.166	0.202	+0.036	21.8%	Warn
PRE14	11.736	11.723	-0.013	0.1%	OK
PRE17	0.074	0.082	+0.008	10.7%	Warn
PRE19	0.004	0.005	+0.001	25.0%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	91.1 ± 1.04
Correlation Stability (*2)	94.0 ± 0.00
Distribution Consistency (*3)	93.4 ± 1.56
Entropy Preservation (*4)	85.2 ± 1.83
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	16.9 ± 2.66

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	69.6 ± 4.22
Feature Accuracy (*10)	83.1 ± 2.55
Novelty Score (*11)	0.7 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	0.8521276595744682	0.8542553191489363	0.006382978723404253	
F1	N/A	0.011764705882352941	0.011764705882352941	0.011764705882352941	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	0.8010638297872345	0.8117021276595745	-0.002127659574468095	
F1	0.11972345964178996	0.04521458933223639	0.09358454106280192	-0.026138918578988018	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	0.8553191489361703	0.85	-0.001063829787234018	
F1	0.037675637056442016	0.05257759784075573	0.013333333333333332	-0.024342303723108676	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8468085106382979	0.8531914893617023	0.8510638297872342	0.004255319148936165	
F1	N/A	0.013333333333333332	0.011764705882352941	0.011764705882352941	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.21063829787234045	0.3244680851063829	0.24255319148936166	0.03191489361702128	
F1	0.256855620661189	0.23067009193223315	0.25366501372694267	-0.003190606934246348	

Model: GB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8382978723404255	0.8329787234042554	0.8319148936170212	-0.006382978723404242	
F1	0.07745532245532245	0.08144393592677346	0.08259803921568627	0.005142716760363819	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8159574468085106	0.7914893617021277	0.8170212765957446	0.0010638297872340385	
F1	0.13088890526971022	0.16818851418022422	0.18626630679262257	0.05537740152291237	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8510638297872342	0.8510638297872340	0.8531914893617023	0.002127659574468066	
F1	N/A	0.009523809523809523	0.01	0.01	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8553191489361703	0	
F1	N/A	0.0	N/A	0	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8553191489361703	0	
F1	N/A	0.0	N/A	0	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.92	1.88	1.0	0.0	0.0	0.0	1.0	1.0	12.0
2	4.0	3.3665473005676323	3.28	1.0	0.0	0.0	0.0	1.0	1.0	12.0
3	3.0	5.555774512982925	62.78865963605349	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	3.3505834931551655	2.56	1.0	0.0	0.0	0.0	1.0	0.0	13.0
5	3.0	2.3052204555353227	1.64	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	2.0	2.8014619830307077	1.6	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.68	1.64	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.08	2.08	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.423072080099651	1.4	0.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	4.082146314943311	2.5918899215002806	2.0	0.0	1.0	0.0	1.0	0.0	12.0
5	2.0	4.216729880324602	2.6126387865459915	2.0	0.0	0.0	0.0	1.0	1.0	11.0
6	3.0	3.781204516535829	2.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.44	2.72	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.88	1.64	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	4.36	2.9577151224051543	0.0	0.0	0.0	1.0	0.0	0.0	11.0
4	3.0	2.8	1.76	1.0	0.0	0.0	0.0	0.0	1.0	12.0
5	3.0	4.16	2.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0
6	2.0	4.36	68.21742549185456	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	4.347748851165552	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	5.0	3.611110888974165	2.84	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.92	2.4	1.0	1.0	0.0	0.0	0.0	0.0	12.0
4	3.0	3.52	2.82	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	4.057234966273288	2.92	1.0	0.0	0.0	0.0	1.0	1.0	11.0
6	3.0	4.2	3.64	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	4.55749154419557 2	2.71207432811154 3	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	3.24	3.18731723972629 45	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	2.0	4.41421612407339 4	4.46525195439368 4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	4.32	2.4	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	3.0	2.48712706533002 95	2.08992738297259 17	1.0	0.0	0.0	0.0	0.0	0.0	12.0
6	3.0	3.67605600034665 8	2.12	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.84	2.52	1.0	0.0	0.0	0.0	0.0	1.0	11.0
2	3.0	2.2	1.72	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.59544693565905 73	1.67854012344885 4	1.0	0.0	0.0	0.0	1.0	1.0	13.0
4	3.0	2.32	1.64	1.0	0.0	0.0	0.0	1.0	1.0	12.0
5	4.0	5.53779110914623 7	4.98364909781753 3	1.0	0.0	0.0	0.0	0.0	0.0	12.0
6	3.0	2.96	2.4	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	1.0	11.0
2	3.0	3.0	2.8	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.4	2.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.76	1.72	1.0	0.0	1.0	0.0	1.0	0.0	12.0
5	3.0	2.4	1.6284838422433912	1.0	0.0	0.0	0.0	1.0	1.0	12.0
6	3.0	2.48	2.48	1.0	0.0	1.0	1.0	1.0	0.0	12.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	5.0	3.76	2.3925687240022726	1.0	0.0	0.0	0.0	0.0	1.0	12.0
2	3.0	2.48	3.12	1.0	0.0	1.0	0.0	0.0	0.0	11.0
3	3.0	3.76	2.56	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	2.0	3.0332106281674465	2.08	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.8288473912438263	2.1111598643039433	1.0	0.0	0.0	0.0	1.0	1.0	12.0
6	3.0	2.48	2.12	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	3.92	4.28935624293657	1.0	0.0	0.0	0.0	0.0	0.0	12.0
2	3.0	2.84	2.36	1.0	0.0	0.0	0.0	0.0	1.0	11.0
3	3.0	4.0	3.52	0.0	0.0	0.0	0.0	1.0	0.0	14.0
4	5.0	2.88	2.15569177904211	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	1.77920711110821	1.36	0.0	0.0	0.0	0.0	0.0	0.0	12.0
6	3.0	2.45430416467777	1.6	0.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.96	2.97209467650136	0.0	0.0	0.0	0.0	1.0	0.0	12.0
2	8.0	3.44	3.2	1.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	4.52	3.92	0.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	2.6	2.08	0.0	0.0	1.0	0.0	0.0	0.0	12.0
5	3.0	2.2	1.68	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	2.0	3.67125008084701	2.84	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0